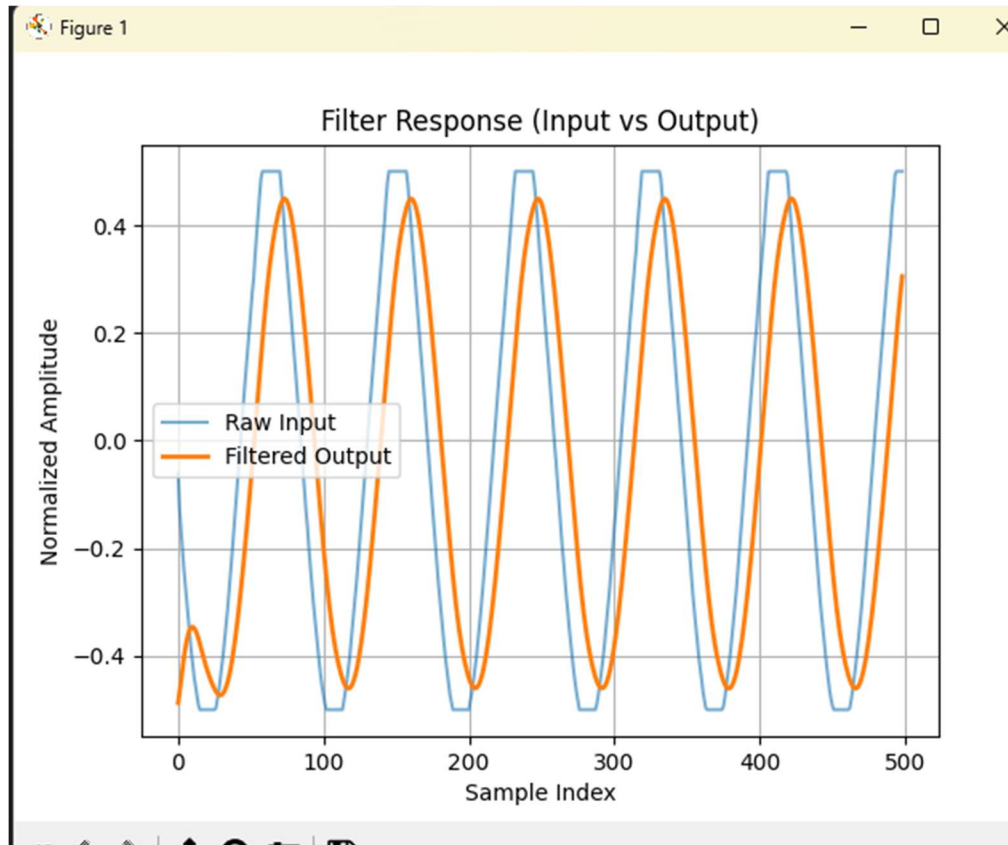


EMA 2nd Order Digital Filter.

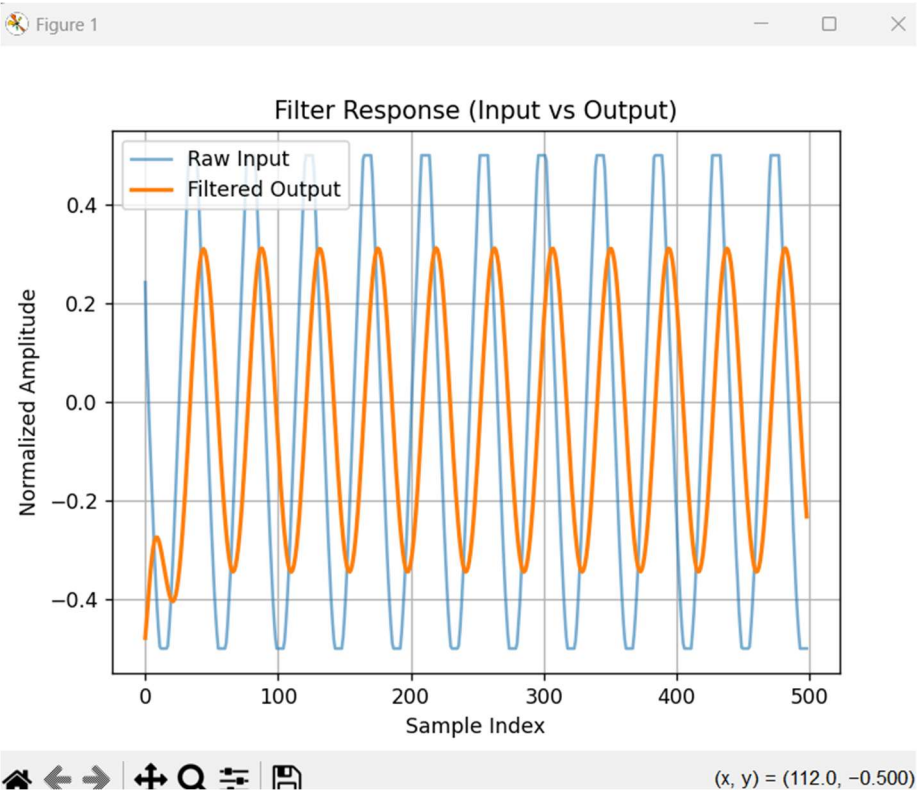
Corresponds to EMA_And_Accel_Logging

Plotted with serialLogAndPlot.py

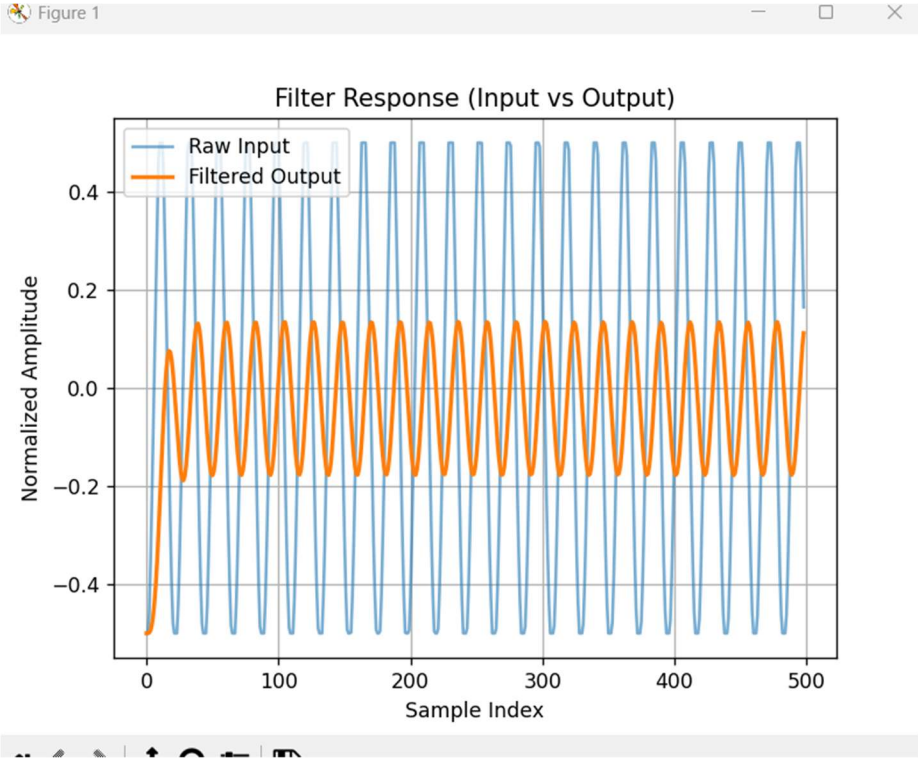
50 Hz input, 2nd ord, 30 Hz cutoff



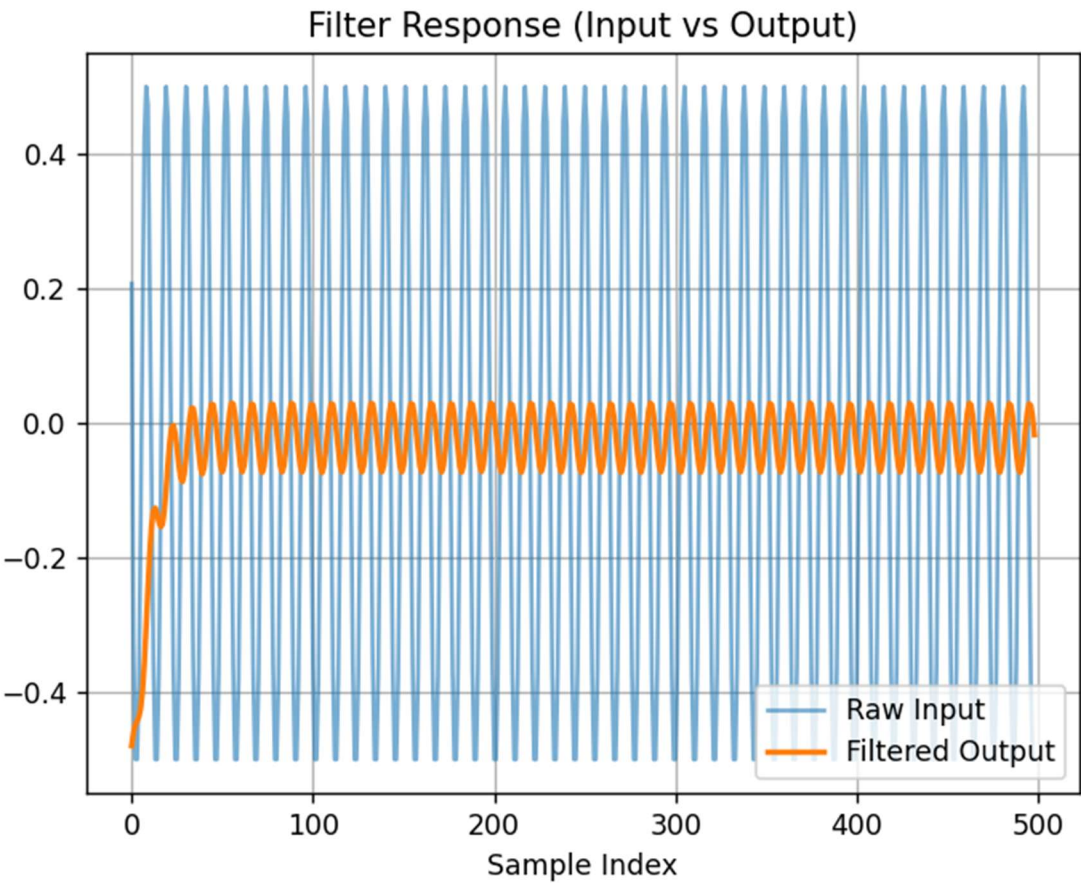
100 Hz



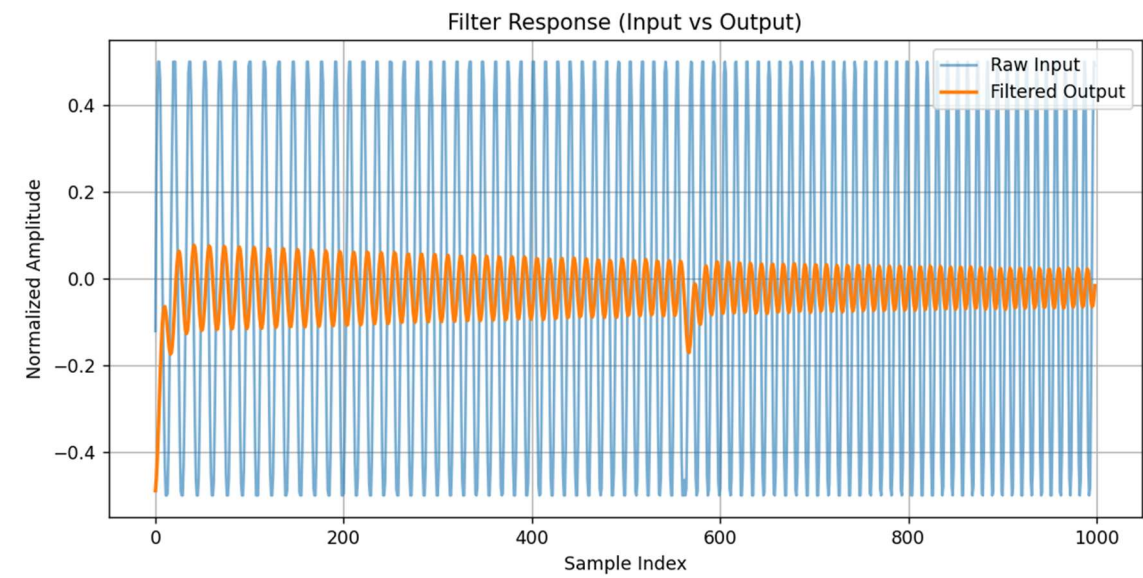
200 Hz



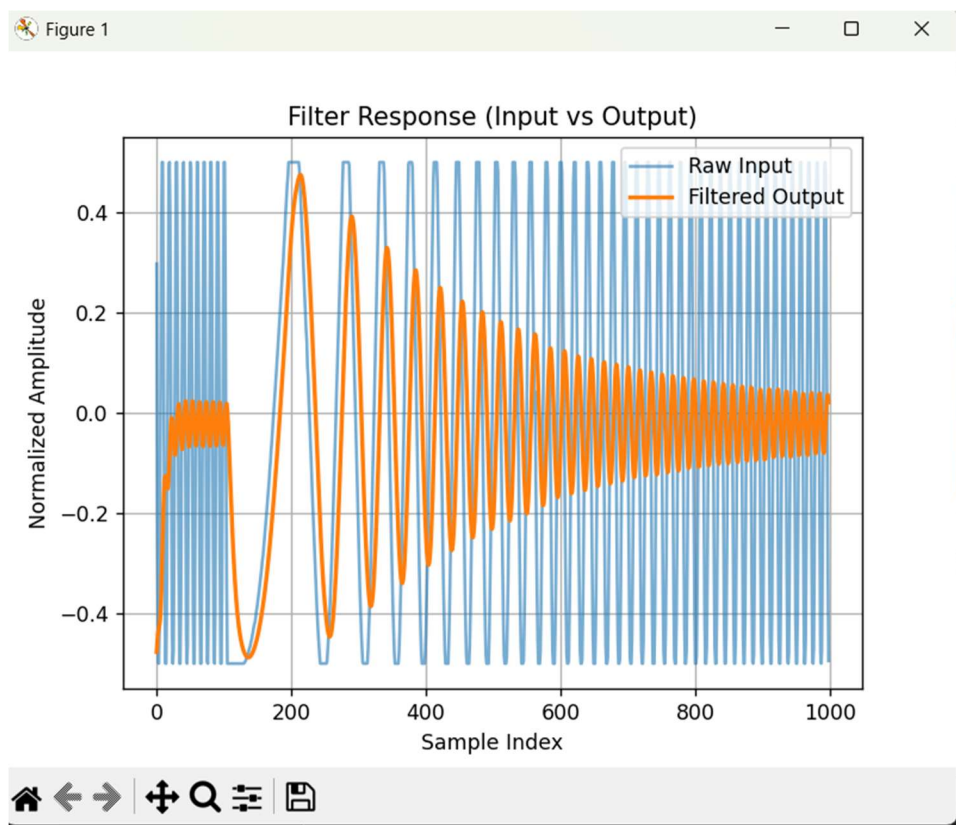
400 Hz



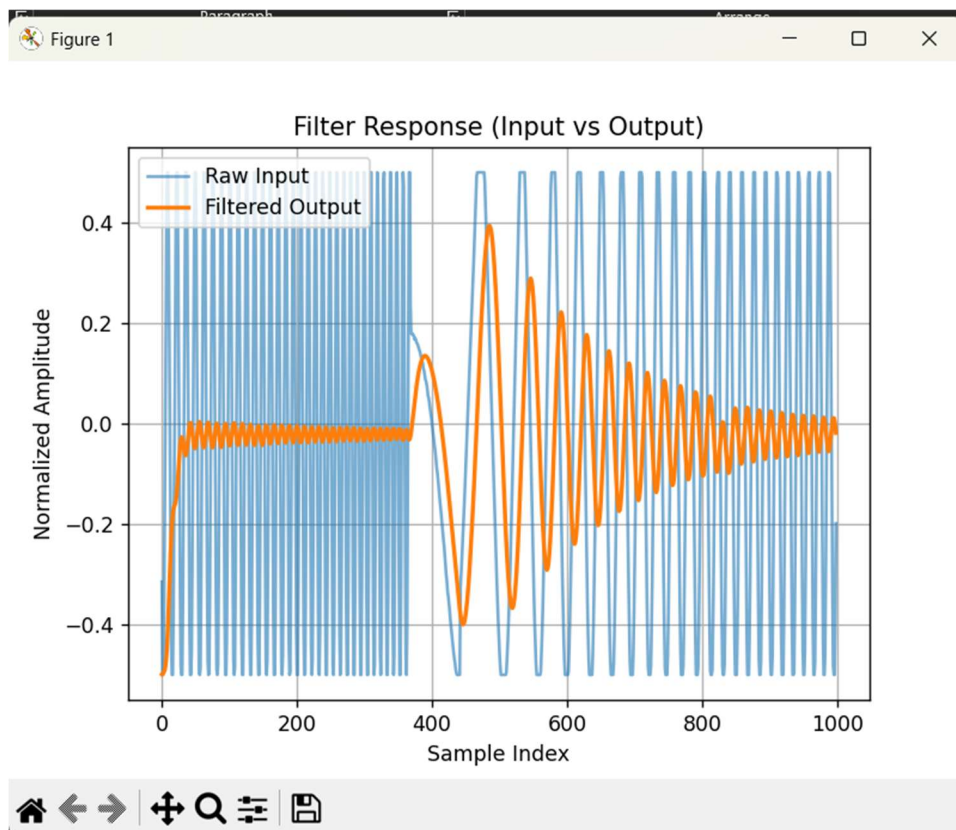
Sweep: 1-450 Hz, 600 ms



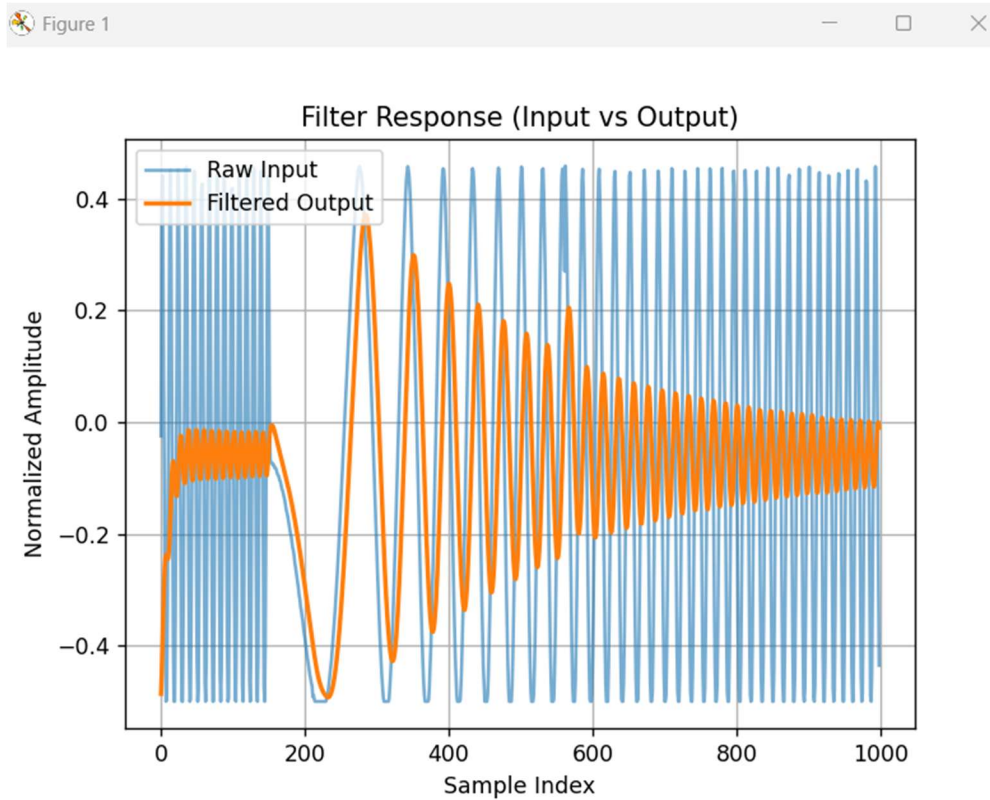
1-450 Hz, 250 ms sweep. 3.3 amp, 1.650 offset



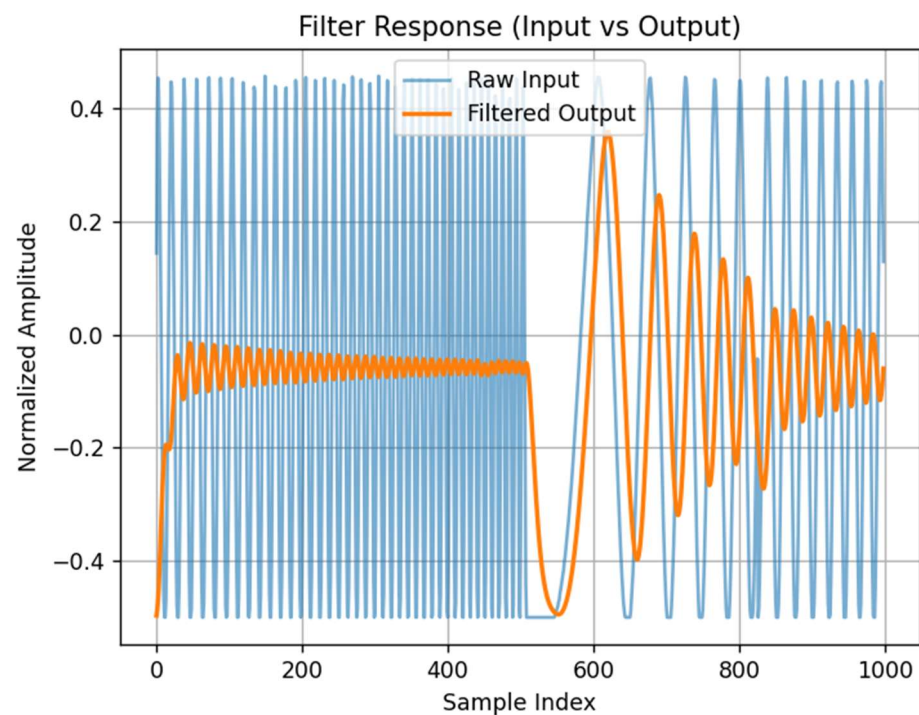
3rd order. 1-450 Hz sweep, 250 ms sweep. 3.3 amp, 1.650 offset



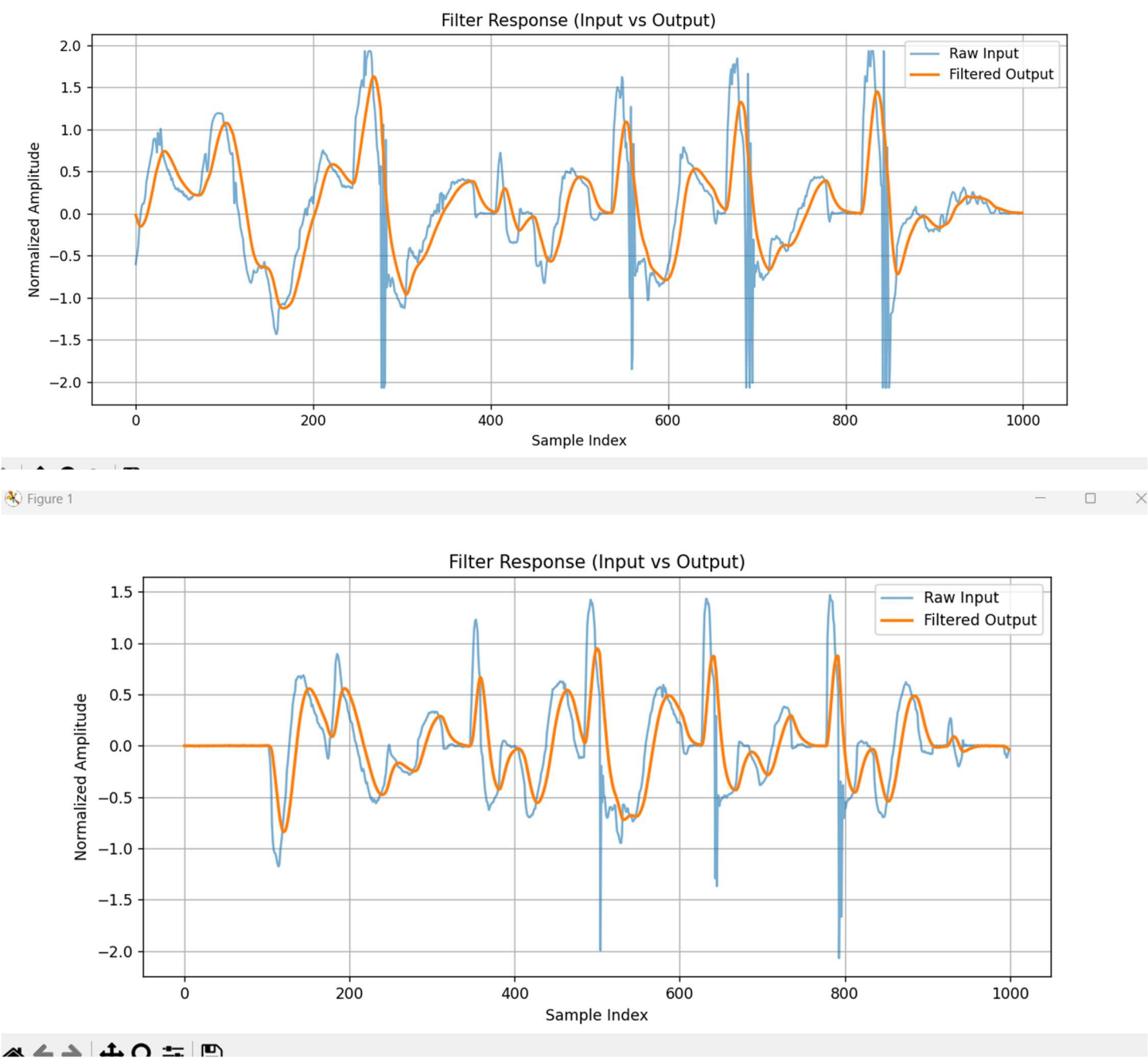
1-450 Hz, 250 ms sweep. 3.0 V amp, 1.550 V offset



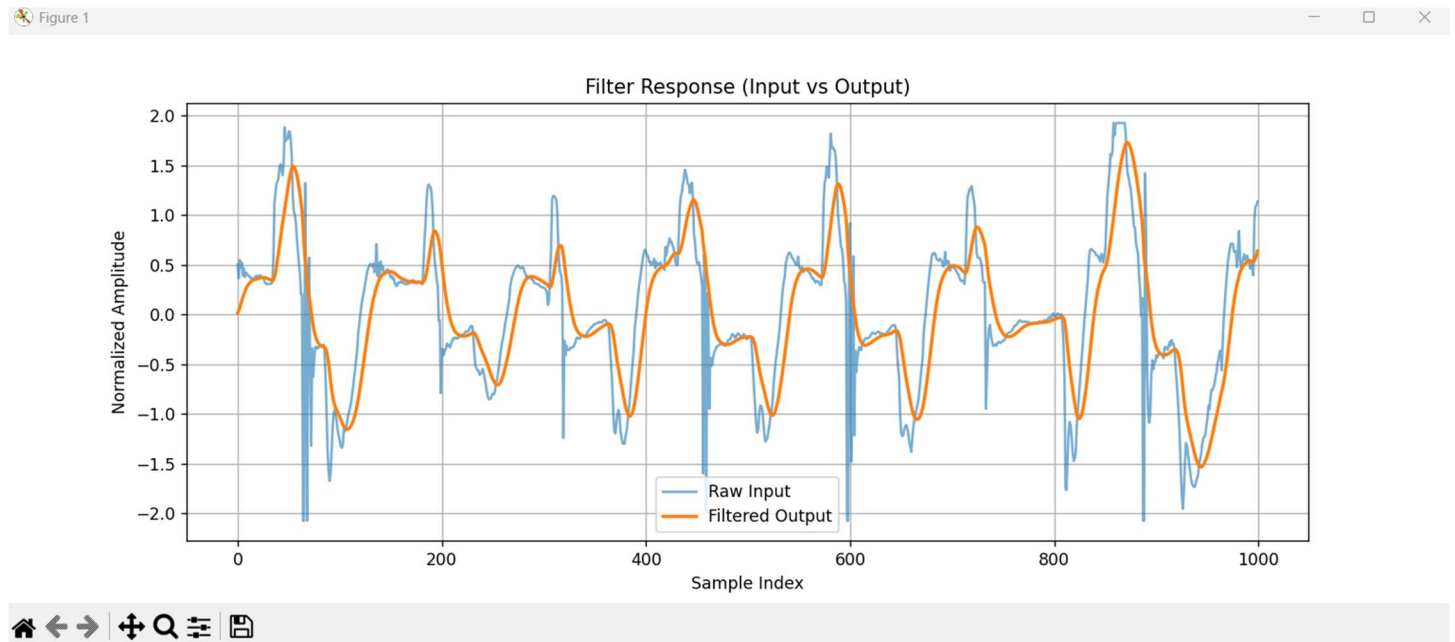
3rd order. 1-450 Hz sweep, 250 ms sweep. 3.0 V amp, 1.550 offset



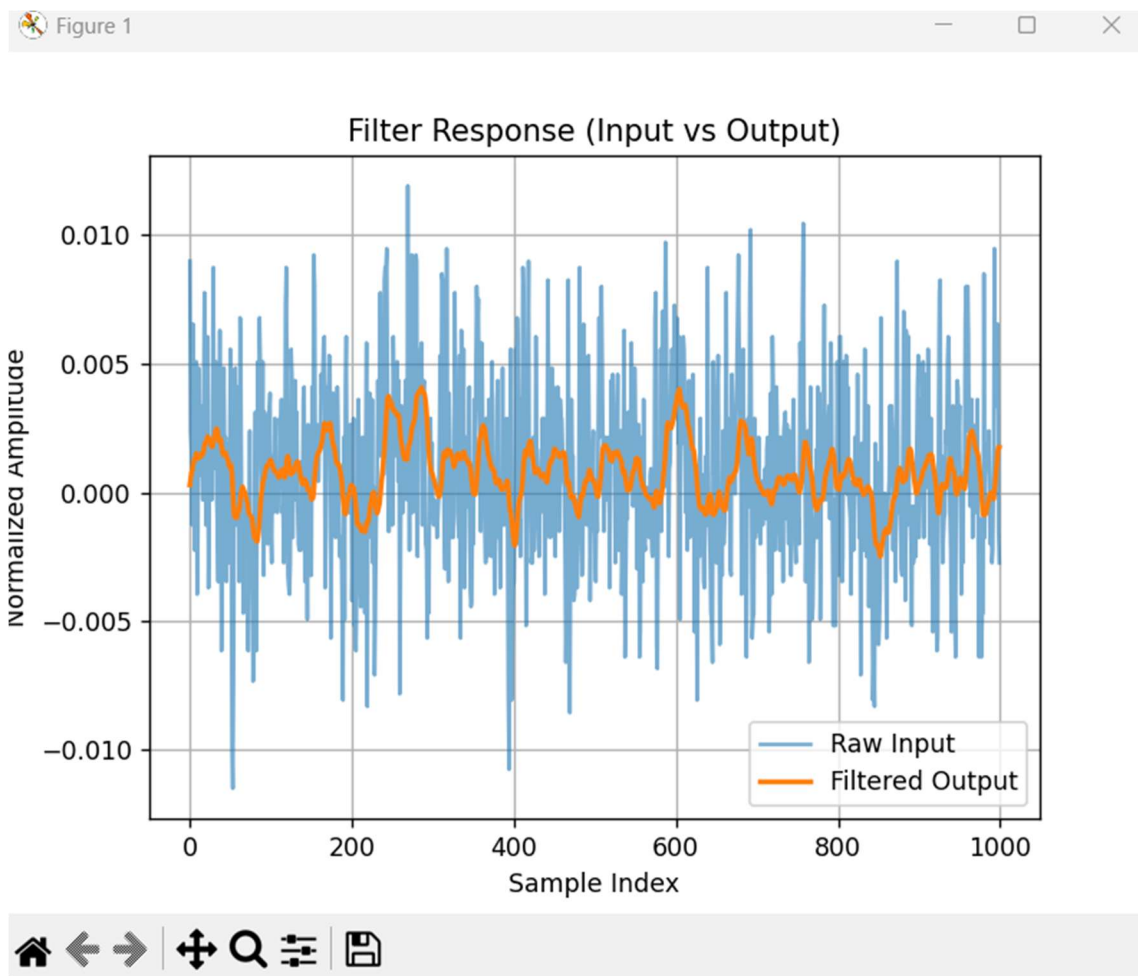
Accelerometer Raw vs filtered accelerometer data (x axis):



Timer Based version of the EMA, cutoff = 31 Hz, second order. Corresponds to EMA_And_Accel_Logging_Timer.py

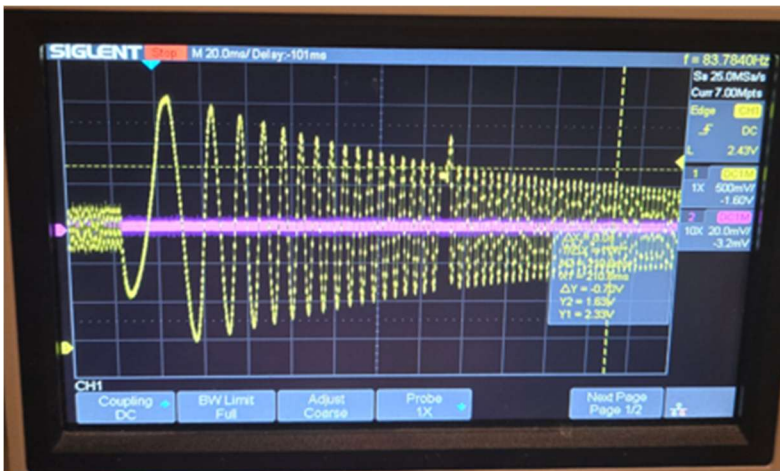


No movement, just noise measurement:

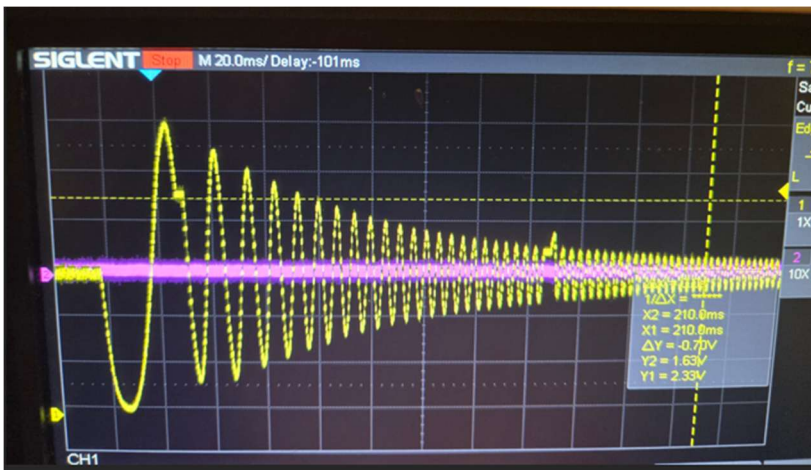


Scope Photos : ADC -> EMA -> DAC -> SCOPE, 3.0 V peak 1-1kHz sweep

1st order



2nd Order



3rd order:

